

SecureVault - Password Manager in C

Mini Project Report

Aadarsh Jha
B.Tech (CSE-AI)
3rd Year

Project Title	SecureVault - Password Manager in C
Project Type	Console Based Application
Programming Language	C
Development Environment	Visual Studio Code, GCC
Version Control	Git & GitHub
Difficulty	Beginner

1. Project Overview

SecureVault is a console-based password manager developed using the C programming language. It allows users to securely manage website credentials using binary file handling. Access to the application is protected by a master password, ensuring that only authenticated users can view or modify stored passwords.

2. Problem Statement

Many users store passwords in notebooks or plain text files, which is insecure and difficult to manage. The aim of this project is to provide a simple password management solution using the core concepts of C Programming.

3. Objectives

- Create and authenticate a Master Password.
- Add new password records.
- Display all saved passwords.
- Display passwords by category.
- Search password records.
- Update existing passwords.
- Delete password records.
- Store data permanently using binary file handling.

4. Features

Master Password Authentication
Add Password
Display All Passwords
Display Passwords by Category
Search Password

Update Password
Delete Password
Change Master Password

5. Modules

Master Password Module

- Create Master Password
- Authenticate User
- Change Master Password

Password Management Module

- Add Password
- Display Passwords
- Search Password
- Update Password
- Delete Password

6. Data Files Used

master.dat - Stores the master password.

vault.dat - Stores all password records in binary format.

7. C Programming Concepts Used

Variables, Arrays, Structures, Functions, Header Files, Loops, Switch Case, Conditional Statements, String Handling, Binary File Handling.

8. Software Requirements

Operating System: Windows/Linux

Compiler: GCC

IDE: Visual Studio Code (Recommended)

9. Future Enhancements

- Password Encryption
- Password Generator
- Password Strength Checker
- Hide Password While Typing
- GUI Version
- Multiple User Accounts
- Backup and Restore

10. Conclusion

SecureVault successfully demonstrates the implementation of a real-world console application using the fundamentals of C Programming. The project makes effective use of structures, functions, modular programming and binary file handling while providing a simple and user-friendly password management system.